

MEDIATEK

MT8676 Android AEE User Manual

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Version History

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1 AEE

1.1 Overview

1.1.1 Brief Introduction

This document provides the guidelines for using the AEE feature to debug exceptions on devices.

1.1.2 Scope

This document is applicable to MediaTek Android system.

1.2 Definitions and Abbreviations

1.2.1 Definition

For the purposes of the present document, the following terms and definitions are applied.

1.2.1.1 AEE

Advanced Exception Engine (AEE) is an exception catching and debugging information generation mechanism. When an exception happens, the device will collect debug information and package into DB (database) files.

1.2.2 Abbreviation

Please note the abbreviations and their explanations are provided in [Table 1-1](#). They are used in many fundamental definitions and explanations in this document and are specific to the information that this document contains.

Table 1-1. Abbreviation

Abbreviation	Description
AEE	Advanced Exception Engine
EE	External Exception
HW Reboot	Hardware Reboot Exception
HWT	Hardware Watchdog Timeout
KE	Kernel Driver Exception
NE	Native Process Exception

1.3 Architecture Overview

1.3.1 DB Type Introduction

AEE generates different types of DB as shown in [Table 1-2](#).

Table 1-2. DB type information

Layer	Type	Level	When DB is Generated?
Java	JE	system server => FATAL Other JE => EXCEPTION	Java Exception
	ANR	Exception	Application no response
	SWT	Fatal	System server watchdog timeout
Native	NE	netd/surfaceflinger => FATAL Other NE => EXCEPTION	Userspace processes receive an exception signal (SIGILL/SIGABRT/SIGBUS/SIGFPE/SIGSEGV).
	System API dump	Exception	Userspace processes call AEE-provided interface.
Kernel	KE	Fatal	Kernel panic occurs.
	HWT		Kernel watchdog timeout occurs.
	HW Reboot		Watchdog timeout occurs and the device hangs for a long time.
	Kernel API dump	Exception	Kernel driver calls AEE-provided interface.
External	EE	Exception	External IC exception occurs.

1.3.2 DB Type Introduction

After exception happens, AEE DB will be stored to /data/aee_exp. Those *.dbg files are DB files.

e.g.

--db.00.SystemAPI

--db.00.SystemAPI.dbg: This is a DB file which has the .dbg suffix.

--ZZ_INTERNAL: This file simply describes the exception.

--db.01.NE

--db.01.NE.dbg

--ZZ_INTERNAL

1.4 Architecture Overview

All AEE configurations are in rc files:

vendor/mediatek/proprietary/external/aee/config_external/init.aee.customer.system.rc

vendor/mediatek/proprietary/external/aee/config_external/init.aee.customer.vendor.rc

1.4.1 How to Enable AEE

AEE feature will be disabled by default on customer side.

If customer needs to enable AEE, please set **MTK_LOG_CUSTOMER_SUPPORT** = yes in following makefile:

```
device/mediateksample/<project>/ProjectConfig.mk
device/mediatek/vendor/<project>/VendorConfig.mk
device/mediatek/system/<project>/SystemConfig.mk
```

1.4.2 Set AEE Mode

By default, AEE mode for customer eng is 3, AEE mode for user/userdebug is 4.

When AEE mode is 3, it can capture both fatal and normal exception DB, but when AEE mode is 4, only fatal exception DB can be captured.

If customers want to switch AEE mode, they need to modify rc file and use `adb shell aee_v2 -m <mode_num>` to set AEE mode.

[How to modify rc file]

Modify `vendor/mediatek/proprietary/external/aee/config_external/init.aee.customer.vendor.rc` by setting the property `ro.vendor.aee.enforcing` to "no". Add the following two lines:

```
```
on init
 setprop ro.vendor.aee.enforcing no
```
```

[how to switch AEE mode by adb command]

```
adb shell aee_v2 -m <mode_num>
```

e.g.

```
adb shell aee_v2 -m 3 --- Switch AEE mode to 3
```

```
adb shell aee_v2 -m 4 --- Switch AEE mode to 4
```

1.4.3 Set Max DB Count

According to DB types described in [Table 1-2](#), the DB level includes Fatal and Normal.

The default max count of Fatal DB is 8 and the default max count of Normal DB is 8.

Users can modify **rc file** to change the max count.

Modify `vendor/mediatek/proprietary/external/aee/config_external/init.aee.customer.vendor.rc` by setting the property .

e.g.

```
on init
    setprop persist.vendor.aee.db.count 15
    setprop persist.vendor.aee.fatal_db.count 15
    setprop persist.vendor.aeev.db.count 15
    setprop persist.vendor.aeev.fatal_db.count 15
```

1.4.4 Switch Native Exception Flow

The MediaTek platform has two different processing procedures for Native Exception.

MTK direct-coredump NE flow:

When a Native Exception occurs, the `aee_core_forwarder` process starts through kernel to collect the coredump file and notifies the AED daemon to generate the NE DB file, which includes `process_coredump`. However, the corresponding Google native debugger mechanism will be disabled.

Google crash_dump/tombstone flow:

If users are accustomed to using the native tombstone mechanism to analyze exceptions, they can switch to the `crash_dump` process by modifying the rc file:

```
vendor/mediatek/proprietary/external/aee/config_external/init.aee.customer.system.rc
on property:persist.vendor.aeev.core.direct=enable
    setprop debug.debuggerd.disable 0
```

1.4.5 AEE UT Test

Java layer (JE):

```
am crash $(pidof com.android.settings | awk '{print $1}')
```

Native layer (NE):

```
kill -11 $(pidof netd | awk '{print $1}')
```

Kernel layer (KE):

```
echo c > /proc/sysrq-trigger
```

External Module (EE):

```
cat /proc/aed/generate-ee (CONFIG_MTK_AEE_UT kernel config should be enabled)
```

1.5 How to Use DB

1.5.1 Pull DB

In the Android system, according to the division between system and vendor, the DB files will be saved in the `/data/aee_exp` or `/data/vendor/aee_exp` directory.

System layer exception DB (JE/NE/SWT/SystemAPI) will be stored in `/data/aee_exp`.

Vendor layer exception DB (KE/HWT/HWR/KernelAPI/EE) will be stored in `/data/vendor/aee_exp`.

1.5.2 Download GAT Tool

Please search GAT on MOL to download the latest version, as shown in [Figure 1-1](#).

Filter Result							
							Download Filter Result
Name	Version	Release Date↓	Category	Product Line	Tool Type	Download	Release Note
GAT	v4.2244.1	2022-12-02	Official	ALPS	Debugging Tools	binary	Release Note

Figure 1-1. Download GAT tool from MediatekOnline

1.5.3 Extract DB

Open MediatekDBViewer in GAT tool. If your PC is running in a Windows operating system, please execute:

```
GAT (Official) _ALPS\GAT_exe_v4.2034.3\gat-win32-x86_64-4.2034.3.c\gat-win32-x86_64-4\tools\MediatekDBViewer.bat.
```

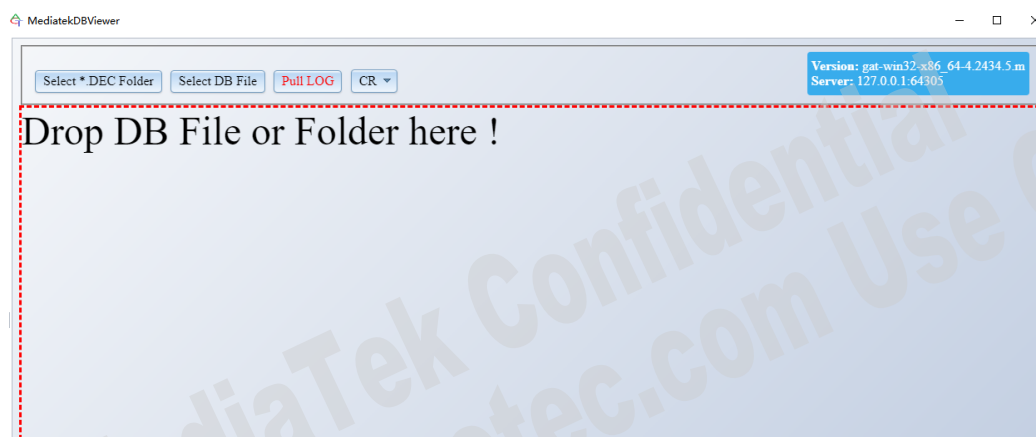


Figure 1-2. MediatekDBViewer tool UI

Click on the “Select DB File” button shown in Figure 1-2, then select the DB file (*.dbg) you want to analyze and click “start” to begin extraction. All the extracted files will be listed on the left side of the UI.

The basic file is __exp_main.txt which describes the basic information of the exception.

1.5.4 Analyze DB Files

1. For NE/KE/HWT issue:

Method 1:

Click “Set Symbols path” and “Analyze” (as shown in Figure 1-3), and then the generated out.json file will show detailed analysis content, including calltrace information.

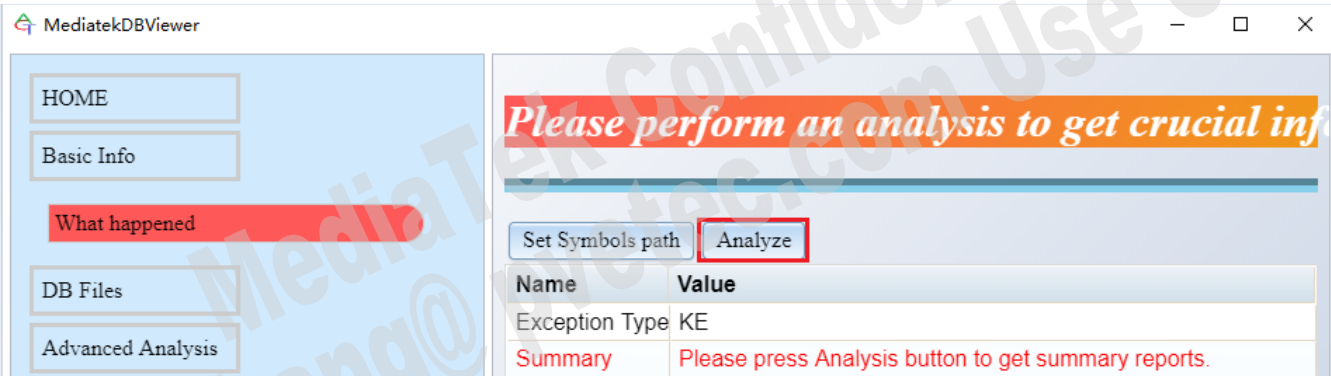


Figure 1-3. Analyze DB

Method 2:

Click “Set Symbols path” and “Launch GDB” (as shown in Figure 1-4), GDB window will show up, then you can get calltrace information by using bt command in GDB.

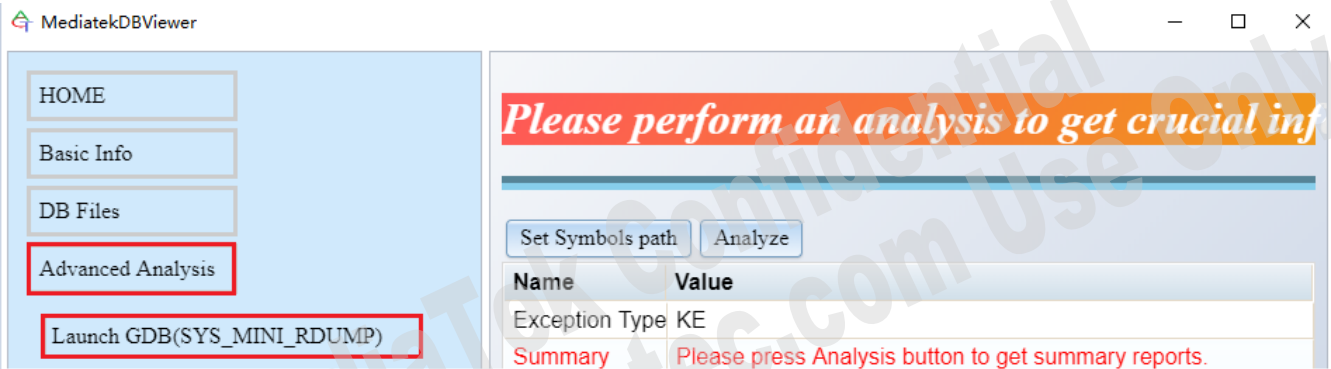


Figure 1-4. Launch GDB

2. For HW_Reboot issue:

HW_Reboot is a special issue: it cannot get backtrace, so you can only check some other debug files in DB (SYS_LAST_CPU_BUS/DFD).

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